Directed Graphs

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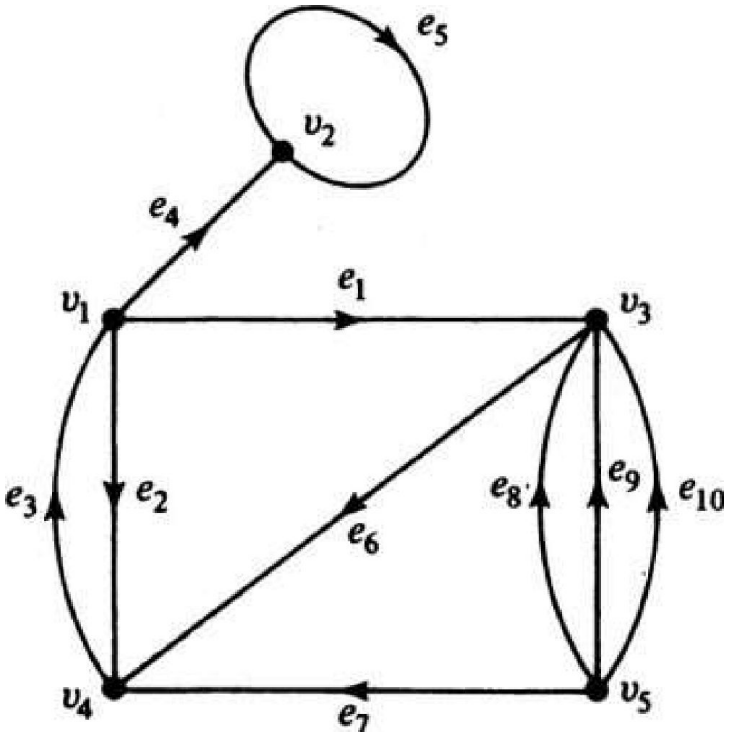
Directed Graphs Definition

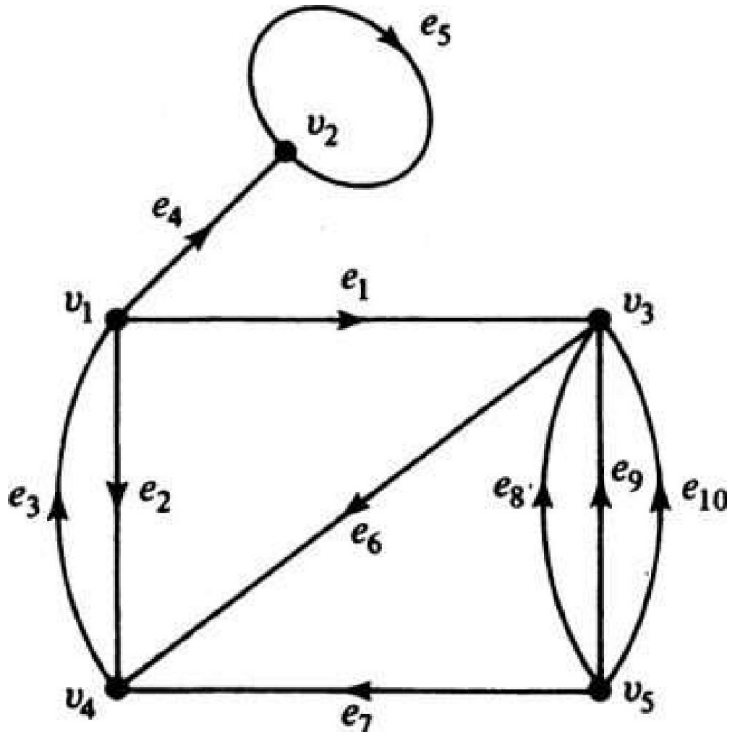
- A **directed graph** (or a **digraph**) **G** consists of a set of **vertices V** = {v_1, v_2, . . .}, a set of **edges E** = {e_1, e_2, . . .}, and a **mapping Ψ** that **maps every edge onto some ordered pair of vertices** (v_i, v_j).
- A **digraph** is also referred to as an **oriented graph**.
- In a digraph, an **edge** is **incident out of** a **vertex** and **incident into** a **vertex**. The **vertex** v_i, which edge e_k is **incident out** of, is called the **initial vertex** of e_k. The vertex v_j, which e_k is **incident into**, is called the **terminal vertex** of e_k.
  - v_5 is the **initial vertex** and v_4 is the **terminal vertex** of **edge** e_7.
  - An **edge** for which the **initial and terminal vertices** are the **same** forms a **self-loop**, such as e_5.



Directed graph **G** with 5 vertices and 10 edges

Directed Graphs Definition

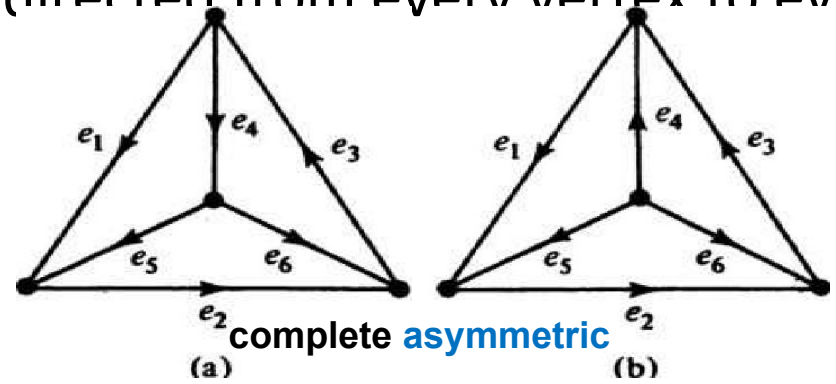
- .The **out-degree** of a **vertex** v_i, $d^+(v_i)$ is defined as the **number of edges incident out** of the vertex v_i.
- The **in-degree** of a **vertex** v_i, $d^-(v_i)$ is defined as the **number of edges** incident into v_i.
- For example,
  - $d^+(v_1) = 3$, $d^-(v_1) = 1$,
  - $d^+(v_2) = 1$, $d^-(v_2) = 2$,
  - $d^+(v_5) = 4$, $d^-(v_5) = 0$
- In any digraph G with n vertices, the sum of all in-degrees is equal to the sum of all out-degrees, each sum being equal to the number of edges in G, i.e.
  - $\sum_{i=1}^{n} d^+(v_i) = \sum_{i=1}^{n} d^-(v_i)$
- For an **isolated vertex** **in-degree** and the **out-degree** are both equal to **zero**.
- For a **pendant vertex** v, $d^+(v) + d^-(v) = 1$



Directed graph G with 5 vertices and 10 edges

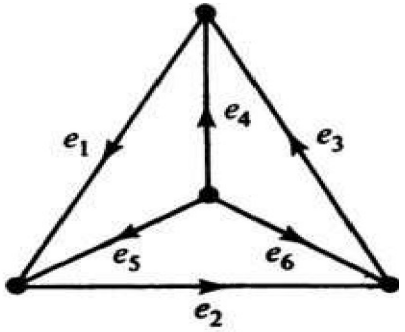
Types of Digraphs

- **Isomorphic diagraph**: For **two digraphs** to be **isomorphic** **not only** must their **corresponding undirected graphs** be **isomorphic**, but the **directions** of the **corresponding edges** must also **agree**.
- **Simple digraphs**: A digraph that has **no self-loop** or **parallel edges** is called a simple digraph
- **Asymmetric digraphs**: Digraphs that have **at most one directed edge between a pair of vertices**, but are allowed to have self-loops, are called asymmetric or antisymmetric.
- **Symmetric digraphs**: Digraphs in which **for every edge** (**a**, **b**) (i.e., from vertex a to b) **there is** also **an edge** (**b**, **a**).
- **Complete digraphs**: A **complete symmetric digraph** has one edge directed from every vertex to every other vertex, and a **complete asymmetric digraph** is an asymmetric digraph in which there is exactly one edge between every pair of vertices.
- **Complete symmetric digraph**: is a simple digraph in which there is exactly one edge directed from every vertex to every other vertex

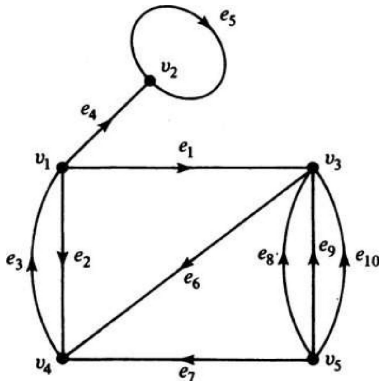


complete asymmetric

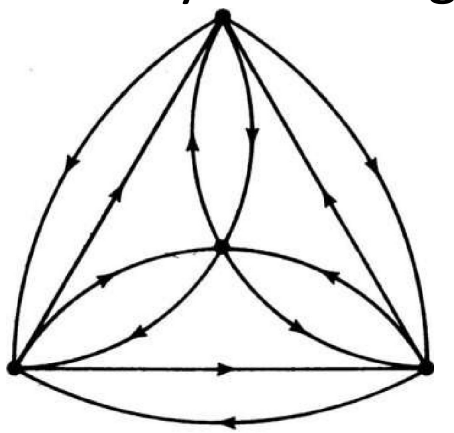
Two non-isomorphic digraphs



Simple digraphs



Asymmetric digraphs

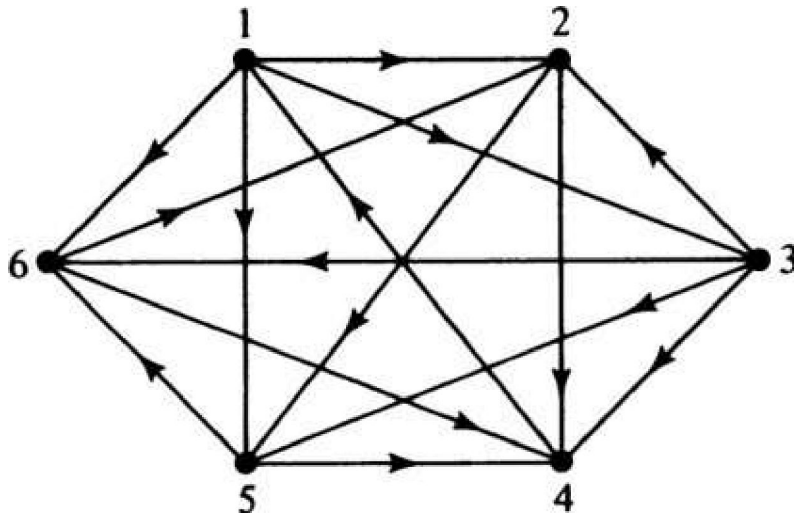


Complete Symmetric digraphs

Paired Comparisons and Tournaments

- A **digraph** is a way of **representing** the **results** of a **paired-comparison experiment**.
- **For example**, the **results** of a **round-robin tournament** in which **every player** has **played against every other** may also be **represented** by a **digraph** in which **an edge directed from vertex a to b represents** the **victory** of **player a over player b**. This **complete asymmetric digraph** is **called** a **tournament**.

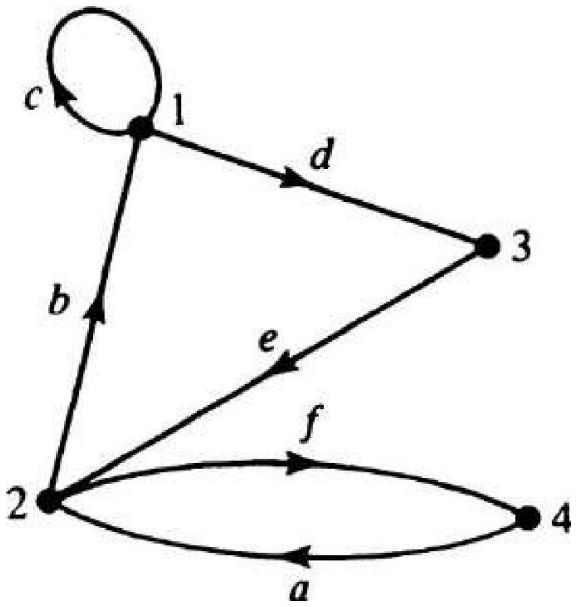
- **Ranking by Score**: A straightforward method of ranking, is to **rank each player** by his/her **score**. The **score** is the **number of games** the player has **won**. The **score** of a **player** in a **tournament equals the out-degree** of the **corresponding vertex** in the digraph.
- If we **use** the **scores for ranking**, we would **rank** the **six player** as (1,3), (2,5,6), and 4. That is, **player 1** and **3** are tied for the **first rank**; there is a three-way tie for the second rank; and **player 4** is the **least one**.



tournament

Euler Digraphs

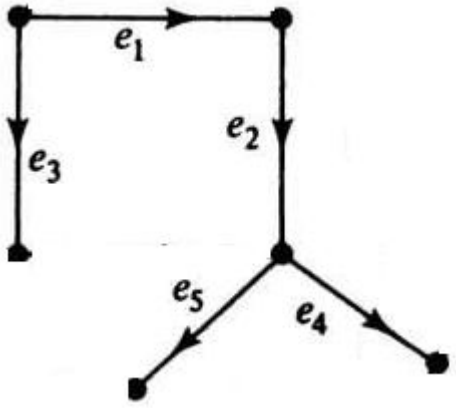
- **Euler digraph**: A **digraph** containing a **directed Euler line** is called an **Euler digraph**. **Note:** a **closed directed walk** (i.e., a directed walk that starts and ends at the same vertex) that **traverses every edge** of the graph **exactly once** is called a **directed Euler line.**
- The graph **G** is an **Euler digraph**, in which the **walk** {**a b c d e f**} is an **Euler line**.
- A **digraph G** is an **Euler digraph if and only if G** is **connected** and is **balanced** [i.e., $d^-(v) = d^+(v)$ for **every vertex v** in **G**]



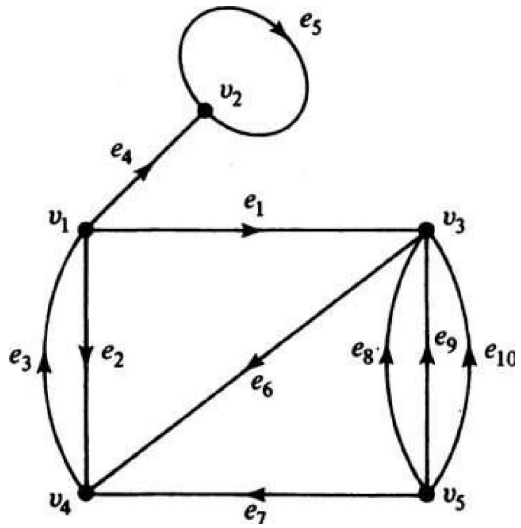
Euler digraph **G**

Trees with Directed Edges

- **Tree**: A **tree** is a **connected digraph** that has **no circuits** (**neither directed circuit nor semicircuit**).
- A **tree** of **n vertices** contains **n – 1 directed edges** and has properties similar to those with undirected edges.
- **Arborescence** is a **rooted tree** with **directed edges** such that
  - **1**. it contains **no circuit** – neither directed nor semicircuit.
  - **2**. there is precisely **one vertex** v of **zero in-degree**.



tree



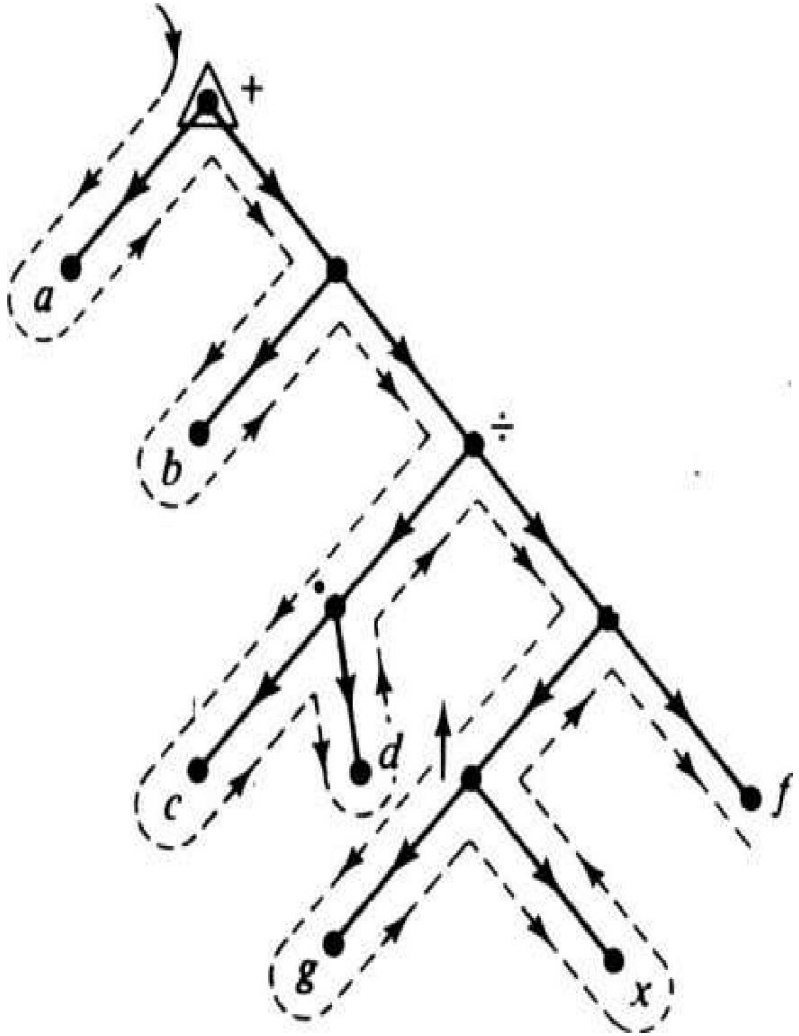
directed circuit: {*e*1, *e*6, *e*3} ;
semicircuit: {*e*1, *e*6, *e*2}.
Both are circuits.

In **arborescence** every **vertex other than** the **root** has an **in-degree** of exactly **one**.

Arborescence

Arborescence Application

- For **evaluating** an **arithmetic expression** the computer must perform the arithmetic operations in a certain order; that **needs** to **scan** the **expression back and forth** to **find** the **sequence of operations** to be **performed**.
- **Arborescence** can be used to **represent an arithmetic expression**, e.g. **a + b – c · d ÷ (g ↑ x – f)** **in the figure**.
- This helps **evaluate the expression** using **Polish notation** with a **single scan** from left to right.
- **arithmetic expression: a + b – c · d ÷ (g ↑ x – f)**
- **In Polish notation: a b + c d · g x ↑ f – ÷ –**



Arborescence for
a + b – c · d ÷ (g ↑ x – f)